

RFP 05

Smart City Analytics Platform

RFP Number

CDA-RFP-2027-005

Issuing Organization

City Development Authority (CDA)

Issue Date

January 2027

Proposal Submission Deadline

15 February 2027

1. Executive Summary

The City Development Authority (CDA) invites qualified vendors to submit proposals for the implementation of a Smart City Analytics Platform.

The objective of this project is to establish a centralized smart city platform capable of collecting, processing, analyzing, and visualizing city-wide operational data from multiple sources, including transportation systems, IoT devices, environmental sensors, and public services.

The platform shall provide real-time insights, predictive analytics, and executive dashboards to support data-driven urban planning and operational excellence.

2. Background

As urban populations continue to grow, city authorities face increasing challenges related to infrastructure management, transportation efficiency, environmental sustainability, and public service delivery.

Current challenges include:

- Traffic congestion

- Limited operational visibility
- Fragmented city data
- Resource allocation inefficiencies
- Environmental monitoring challenges

The Authority seeks a centralized platform that enables proactive decision-making and supports future smart city initiatives.

3. Project Objectives

Operational Objectives

- Improve city operations.
- Optimize resource allocation.
- Improve transportation management.
- Enhance infrastructure monitoring.

Strategic Objectives

- Support smart city transformation.
- Improve citizen experience.
- Enable predictive urban planning.
- Improve sustainability initiatives.

Technology Objectives

- Integrate IoT data sources.
 - Provide advanced analytics.
 - Enable real-time monitoring.
 - Deliver executive dashboards.
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4. Scope of Work

4.1 Smart City Data Platform

The platform shall provide:

- Centralized Data Collection
- IoT Integration
- Data Storage
- Data Processing
- Data Governance

4.2 Traffic Analytics

The solution shall support:

- Traffic Monitoring
 - Congestion Detection
 - Traffic Forecasting
 - Incident Detection
 - Traffic Performance Reporting
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4.3 Environmental Monitoring

The platform shall provide:

- Air Quality Monitoring
 - Noise Monitoring
 - Weather Analytics
 - Environmental Dashboards
 - Sustainability Reporting
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4.4 Public Transportation Analytics

The platform shall support:

- Fleet Monitoring
 - Route Optimization
 - Passenger Analytics
 - Service Performance Monitoring
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4.5 Predictive Analytics

The solution shall provide:

- Demand Forecasting
 - Infrastructure Planning
 - Resource Optimization
 - Predictive Maintenance
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4.6 GIS Analytics

The platform shall provide:

- Interactive Maps
 - Geospatial Analytics
 - Location Intelligence
 - Spatial Reporting
 - Asset Mapping
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5. Technical Requirements

Platform Requirements

The solution shall be:

- Cloud Native
 - Highly Available
 - API Enabled
 - Scalable
 - Mobile Accessible
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Integration Requirements

The platform shall integrate with:

- IoT Sensors
 - Traffic Cameras
 - GPS Systems
 - Environmental Sensors
 - Public Transportation Systems
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Security Requirements

The solution shall support:

- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Audit Logging
- Data Encryption
- Security Monitoring

6. Non-Functional Requirements

Availability

99.95% Availability

Scalability

Support Millions of Daily Events

Performance

Near Real-Time Processing

Reliability

Disaster Recovery Support

7. Deliverables

1. Smart City Analytics Platform
 2. IoT Integration Layer
 3. GIS Analytics Module
 4. Executive Dashboard
 5. Technical Documentation
 6. User Training
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8. Project Timeline

Phase	Duration
Assessment	1 Month
Design	1 Month
Development	4 Months
Integration	2 Months
Testing	1 Month

Phase	Duration
Deployment	1 Month

Total Duration: 10 Months

9. Financial Requirements

Estimated Budget:

SAR 5,000,000

10. Evaluation Criteria

Criteria	Weight
Technical Proposal	50%
Smart City Experience	20%
Team Qualifications	10%
Financial Proposal	20%